

Smart Traffic Lights Using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers and cannot easily react when one road suddenly becomes crowded. This paper presents the concept of smart traffic lights using local artificial intelligence (AI), where traffic information is processed closer to the road. Local processing can reduce dependence on continuous internet communication and may allow faster traffic-control decisions. The proposed system also considers emergency vehicle detection as an important research direction.

Index Terms—Smart Traffic Lights, Local AI, Artificial Intelligence, Traffic Management, Machine Vision, Emergency Vehicle Detection

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate traffic conditions and then adjust the lights. Instead of changing lights only according to a clock, the system uses information collected from the road. Research reviews also show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

II. LITERATURE REVIEW

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay.

Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions. A small local computer can examine camera input and send a short instruction to the traffic-light controller.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

III. SYSTEM COMPARISON

The following table presents the comparison data specified in the workshop answer key. Lower decision time means a faster decision.

Basic timed lights use fixed timing and generally do not require internet data. Cloud AI systems send traffic information to a central server for processing. Local smart-camera

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision speed (s)	Internet use	Setup cost (\$)
Basic Timed Lights	30	Zero	200
Cloud AI Systems	2	High	2,500
Local Smart Cameras	0.5	Low	1,200

systems process selected traffic information near the road. The listed values are the workshop data and are not presented as measurements independently obtained in this paper.

IV. SYSTEM ARCHITECTURE

The proposed architecture follows four main steps. First, Street Cameras capture traffic conditions. Second, the video information is sent to Smart Pole Computers, which perform local processing and make a quick traffic decision. Third, selected information can be forwarded to the City Cloud Server for storage and wider analysis. Fourth, the decision is sent to the Traffic Lights, which change their signals.

The overall flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

V. ARCHITECTURE DIAGRAM

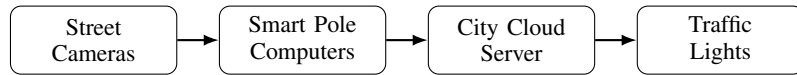


Fig. 1. Proposed Smart Traffic-Light Architecture

VI. RESEARCH GAP

Existing traffic systems work well for regular daily traffic, but they lack a built-in way to automatically clear a path for a speeding ambulance during an emergency. This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route. Emergency-vehicle coordination can therefore be considered a potential direction for further research.

VII. POTENTIAL ADVANTAGES

The proposed local-AI approach can provide several potential advantages:

- Faster local processing of traffic information.
- Reduced dependence on continuous internet communication.
- Better adaptation to changing traffic conditions.
- Possibility of detecting emergency vehicles.
- Reduced amount of raw video data sent to the cloud.

VIII. FUTURE SCOPE

Future research can focus on improving vehicle detection, emergency vehicle recognition, traffic-density estimation, and coordination of multiple nearby traffic signals. Further testing with real traffic data can also be used to evaluate the performance of local AI systems under different road conditions.

IX. CONCLUSION

Smart traffic lights using local AI provide a possible approach for making traffic signals more responsive to changing road conditions. Processing information near the road can reduce dependence on cloud communication and support faster decisions. The integration of emergency vehicle detection is an important direction for developing a more responsive traffic-management system.

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